Introduction to Probability and Statistics

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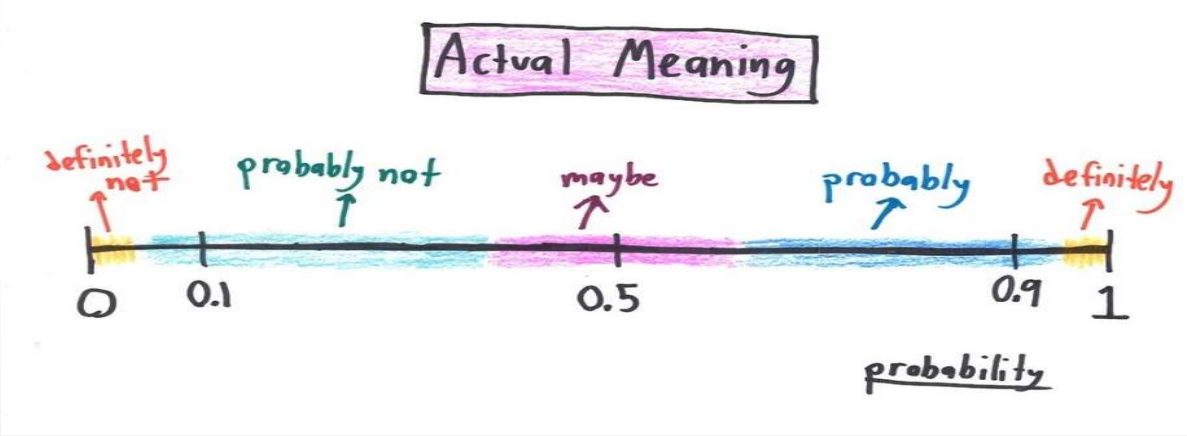
Today's Agenda

- Introductions
- Get to know the course
- Some basic probabilistic frameworks

Introductions

- Who am I?
  - Bachelor's in Mathematics
  - Master's in Statistics
  - Research Interests/Experience: Bayesian Model Averaging, Statistical Inference, Ensemble Climate Projections
- Who are you?
  - Your name
  - Some interesting fact about you (For example: a hobby you have)

Kahoot



https://create.kahoot.it/details/math-probability-warm-up/6201702f-fd2b-4540-a953-46c1041c1523

- Why ‘Study of Probability’ is important now?

Science & Engineering together have developed complex systems that inherently contain uncertainties

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Role of information using data analysis has an important role in making decisions

Context for this course

- As engineers and scientists, we design and modify ***complex systems***.
- The design or modification of any complex system involves the making of choices from various feasible alternatives.
- These choices are made based on criteria such as:
  - Cost
  - Reliability
  - Performance
- How to evaluate these criteria for various alternatives?
  - Actual implementation and experimental evaluation of these alternatives
  - **Utilize "Models" of these alternatives**

Models

- We need to able to understand, model, analyze & predict uncertainties & have a grip on the mathematical models, tools and procedures for determining the inherent randomness in complex systems
- A model is an approximate representation of a situation
- A useful model explains all the relevant aspects of a given situation
- A model can be used instead of actual implementation and experimentation to answer questions about the given situation.
- A mathematical model employs mathematical constructs to represent a situation.

Examples of Models

Deterministic Situations

- Converting a temperature from Fahrenheit to Celsius
- The value of a savings account after 3 months if the interest rate is fixed

Uncertain Situations

- Flipping a coin
- Daily change in the price of a stock
- Wickets taken by Shaheen Shah Afridi during the first over of a T20 match
- Number of students in a lecture of this course

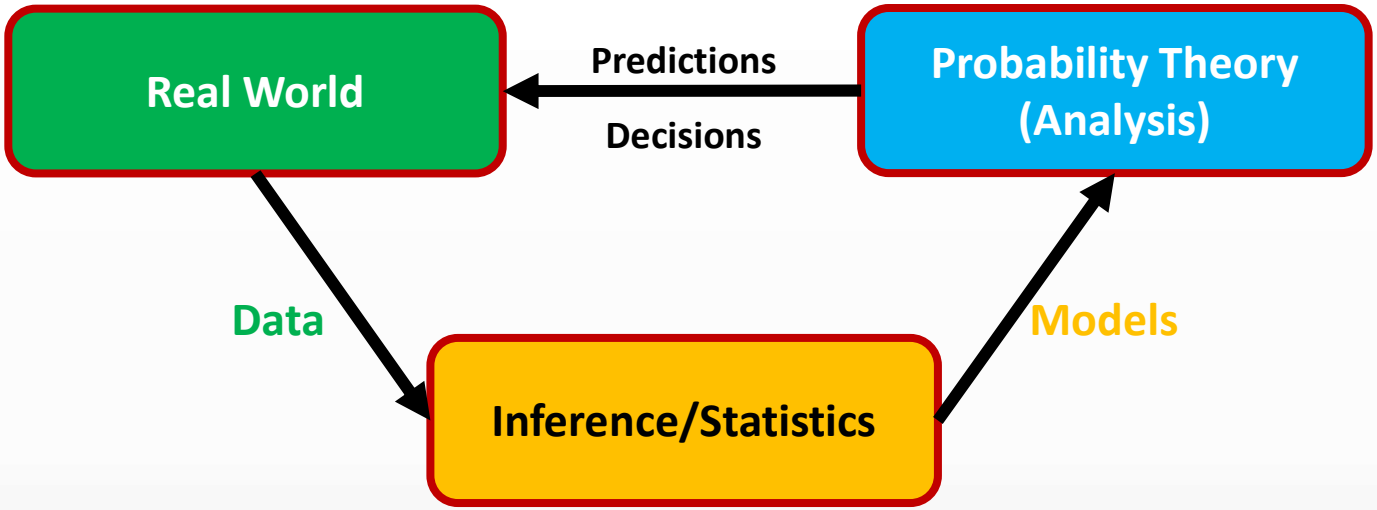
- Question No I: What is the probability of getting a Head in a throw of an un-biased coin?
- Question No II: What is the probability of getting n-heads in n throws of a un-biased coin?
- Question No III: How many times should this biased coin be thrown that there is a better than 0.5 chance (probability) of obtaining at least one head?

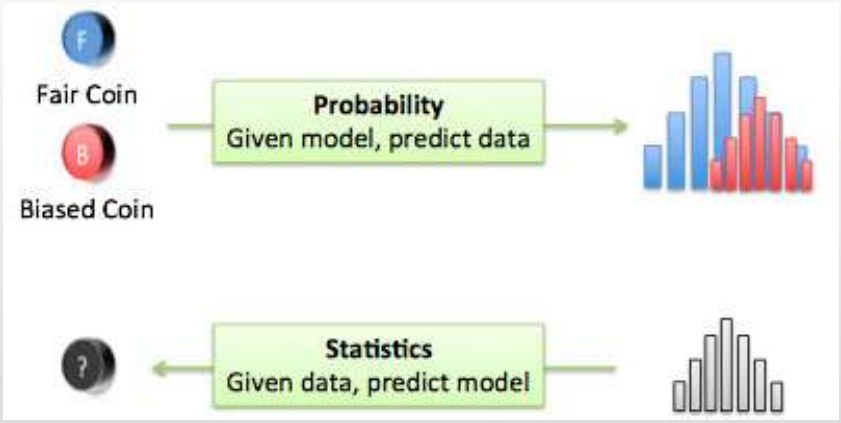
- https://flipsimu.com/



Purpose of this Course

- Course will cover foundations of probability and statistics, with examples from electrical engineering, computer science, and everyday life.
- Course contents can broadly be divided:-
    1. **fundamentals of probability**
    2. **common probability models, and**
    3. **statistics**
- Objectives:-
    - Gaining an appreciation for the limitation of human intuition when faced with finding optimal solution for a problem in presence of uncertainty
    - Recognizing presence of uncertainty in a scenario and modeling it using formal notions of probability
    - Identifying the right questions to ask when drawing inferences from a probability model and interpreting results correctly, develop the skill of probabilistic reasoning.

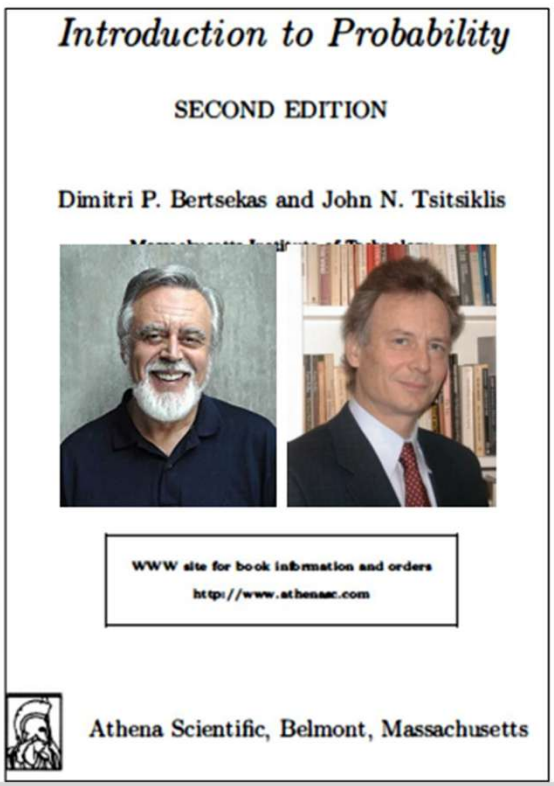

Real World
Predictions
Decisions
Probability Theory (Analysis)
Data
Models
Inference/Statistics

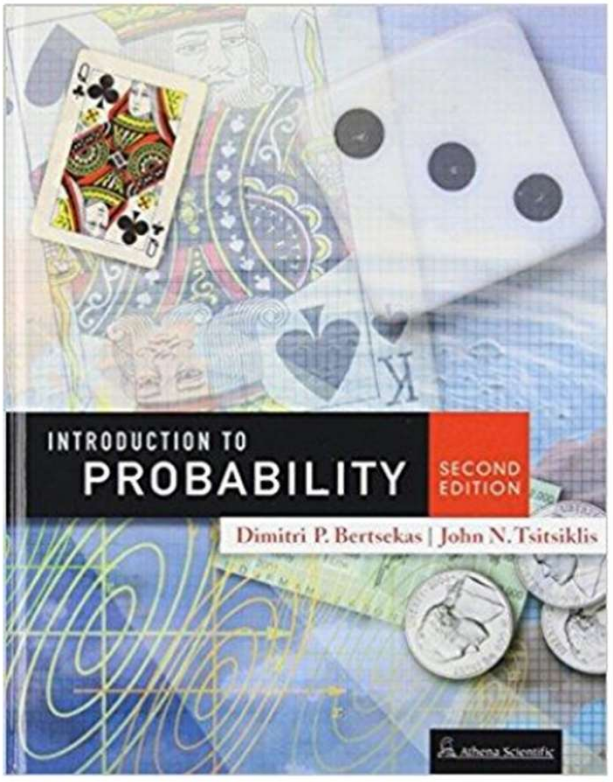

Fair Coin
Biased Coin
Probability
Given model, predict data
Statistics
Given data, predict model

Syllabus

Required Textbook

- Bertsekas, Dimitri P., and John N. Tsitsiklis. **Introduction to Probability.** Belmont, MA: Athena Scientific, 2002





Structure of Required Course Reading

Chapter **Title**

1. **Sample Space and Probability**
   - Sets, Probabilistic Models, Conditional Probability, Total Probability, Theorem and Bayes' Rule, Independence, Counting

2. **Discrete Random Variables**
   - Basic Concepts, Probability Mass Functions, Functions of Random Variables, Expectation, Mean, and Variance, Joint PMFs of Multiple Random Variables, Conditioning, Independence

3. **General Random Variables**
   - Continuous Random Variables and PDFs, Cumulative Distribution Functions, Normal Random Variables, Joint PDFs of Multiple Random Variables, Conditioning, The Continuous Bayes' Rule

4. **Further Topics on Random Variables**
   - Derived Distributions, Covariance and Correlation, Conditional Expectation and Variance Revisited, Transforms, Sum of a Random Number of Independent Random Variables

Structure of Required Course Reading

Chapter **Title**

5. **Limit Theorems**
   - Markov and Chebyshev Inequalities, The Weak Law of Large Numbers, Convergence in Probability, The Central Limit Theorem, The Strong Law of Large Numbers
6. **The Bernoulli and Poisson Processes**
   - The Bernoulli Process, The Poisson Process
7. **Markov Chains**
   - Discrete-Time Markov Chains, Classification of States, Steady-State Behavior, Absorption Probabilities and Expected Time to Absorption, Continuous-Time Markov Chains
8. **Bayesian Statistical Inference**
   - Bayesian Inference and the Posterior Distribution, Point Estimation, Hypothesis Testing, and the MAP Rule, Bayesian Least Mean Squares Estimation
9. **Classical Statistical Inference**
   - Classical Parameter Estimation, Linear Regression, Binary Hypothesis Testing, Significance Testing

Tentative course schedule – Ambitious
Do Not Worry!

"In each field, we must carefully distinguish three aspects of the theory:

(a) the formal logical content,

(b) the intuitive background, and

(c) the applications.

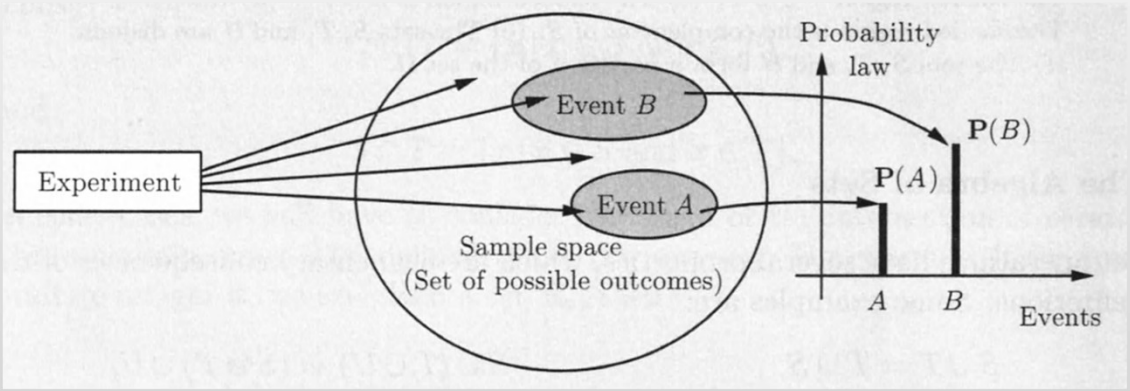
The character, and the charm, of the whole structure cannot be appreciated without considering all three aspects in their proper relation."

- William Feller (An Introduction to Probability Theory and its Applications, 1950)

Probabilistic Model

Every probabilistic model involves an underlying process, called the ***Experiment,*** that will produce exactly one out of several possible outcomes.

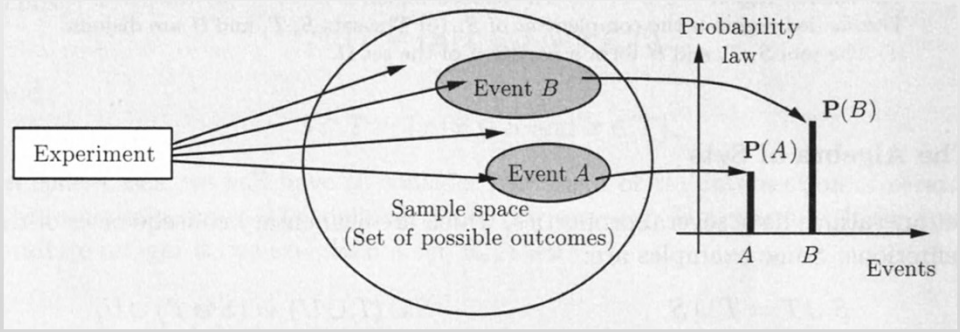
- Sample Space (Ω)
  - The set of all possible outcomes of an experiment
- Probability Law
  - Assigns to a set A of possible outcomes a non-negative number P(A) that encodes our belief about the collective "likelihood" of the elements of A.



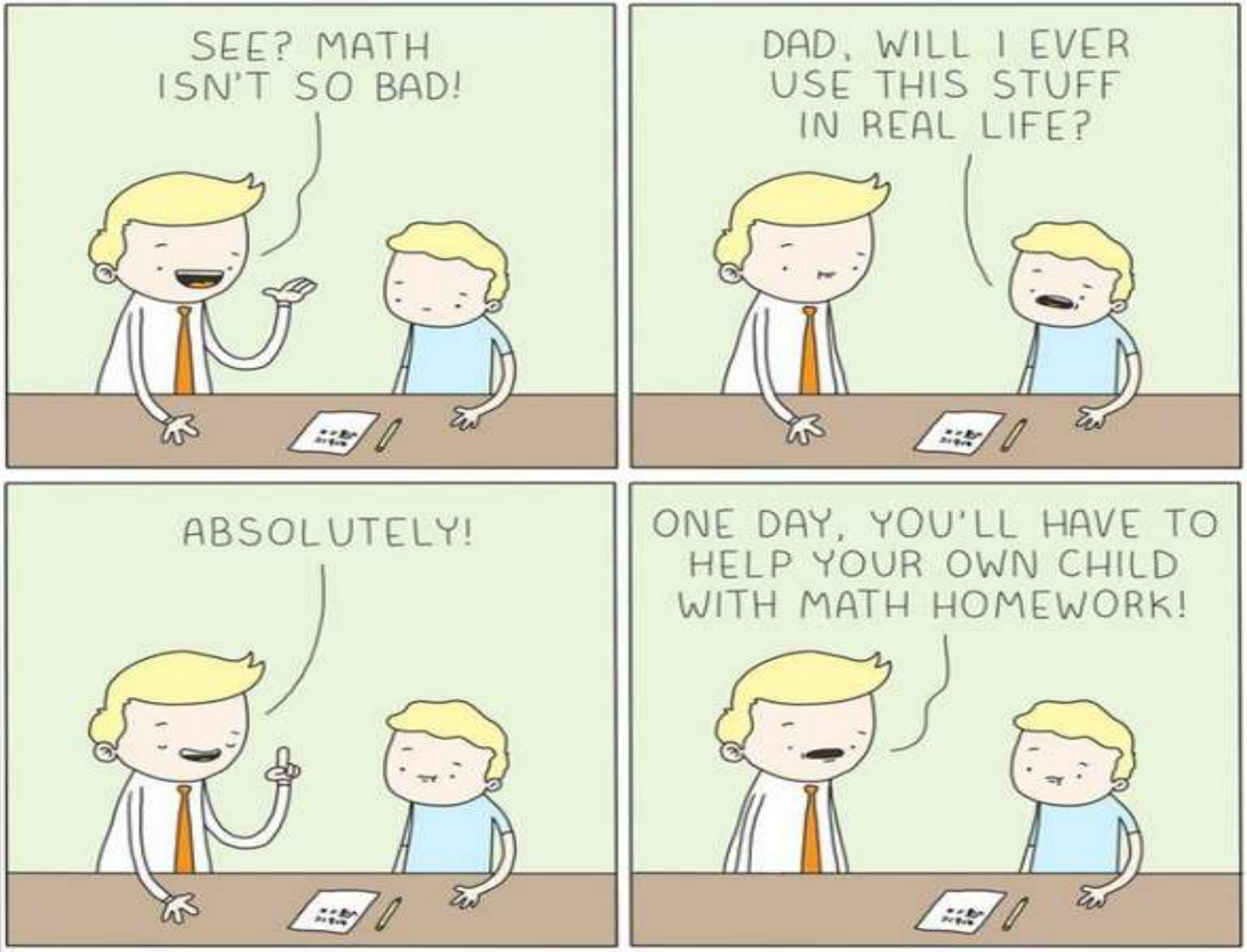
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- Probability Law
  - Assigns to a set A of possible outcomes a non-negative number P(A) that encodes our belief about the collective "likelihood" of the elements of A.
  - This probability law must satisfy certain properties, known as "Probability Axioms."



1. $P(A) \geq 0$
2. $P(A \cup B) = P(A) + P(B)$
3. $P(\Omega) = 1$


SEE? MATH ISN'T SO BAD!
DAD, WILL I EVER USE THIS STUFF IN REAL LIFE?
ABSOLUTELY!
ONE DAY, YOU'LL HAVE TO HELP YOUR OWN CHILD WITH MATH HOMEWORK!